

EM algorithm for the bivariate Poisson

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Here we follow the simplest bivariate Poisson in that paper. That is,

$$X_i = W_{1i} + W_{3i} \quad (1)$$

$$Y_i = W_{2i} + W_{3i} \quad (2)$$

Here W_{ji} are independent and follow Poisson distribution with parameters (λ_{ji}) . (X_i, Y_i) is the goals from home and away teams in match i .

$$\log(\lambda_{1i}) = \mu + home + att_{h(i)} + def_{g(i)} \quad (3)$$

$$\log(\lambda_{2i}) = \mu + att_{g(i)} + def_{h(i)} \quad (4)$$

$$\log(\lambda_{3i}) = \beta_{h(i)}^{home} \quad (5)$$

Here $h(i)$ and $g(i)$ are known deterministic functions assigning the match index to the home and away teams. For identifiability, we further set

$$\sum def_j = 0 \quad (6)$$

$$\sum att_j = 0 \quad (7)$$

Here we do not use β^{common} as in the paper so $\sum \beta_j^{home} = 0$ is not needed anymore. Other assumptions are the same as that paper. And we denote all parameters to be estimated as θ and all results of each match as data.

1 E stage

In E state we seek for the Q function (suppose some iterated $\theta^{(n)}$).

$$E_{W_3|\text{data}, \theta^{(n)}}[\log(P(\text{data}, W_3|\theta))]$$

Firstly, for each i , we express the first conditional probability $P(W_{3i} | (X_i, Y_i), \theta^{(n)})$ by Bayes formula

$$P(W_{3i} = \omega | (X_i, Y_i) = (x_i, y_i), \theta^{(n)}) = \frac{P(W_{3i} = \omega, (X_i, Y_i) = (x_i, y_i) | \theta^{(n)})}{P((X_i, Y_i) = (x_i, y_i) | \theta^{(n)})}$$

Here $\omega \in \{0, 1, \dots, \min(x_i, y_i)\}$. Therefore,

$$\begin{aligned} P(W_{3i} | (X_i, Y_i), \theta^{(n)}) &\propto P(W_{3i} = \omega, W_{1i} = x_i - \omega, W_{2i} = y_i - \omega | \theta^{(n)}) \\ &= \frac{(\lambda_{1i}^{(n)})^{(x_i - \omega)}}{(x_i - \omega)!} \frac{(\lambda_{2i}^{(n)})^{(y_i - \omega)}}{(y_i - \omega)!} \frac{(\lambda_{3i}^{(n)})^\omega}{\omega!} \end{aligned} \quad (8)$$

The normalization is just the probability of $P((X_i, Y_i) = (x_i, y_i) | \theta^{(n)})$ where $(X_i, Y_i) \sim BP(\lambda_{1i}^{(n)}, \lambda_{2i}^{(n)}, \lambda_{3i}^{(n)})$. Denote

$$p_i(\omega | \theta^{(n)}) := P(W_{3i} | (X_i, Y_i), \theta^{(n)})$$

Secondly,

$$\begin{aligned} \log(P((X_i, Y_i) = (x_i, y_i), W_{3i} = \omega | \theta)) &= \log(P(W_{3i} = \omega, W_{1i} = x_i - \omega, W_{2i} = y_i - \omega | \theta)) \\ &= (x_i - \omega) \log \lambda_{1i} - \lambda_{1i} + (y_i - \omega) \log \lambda_{2i} - \lambda_{2i} + \omega \log \lambda_{3i} - \lambda_{3i} + C \end{aligned} \quad (9)$$

Denote the last formula as $l_i(\omega, \theta)$.

Thus,

$$Q(\theta | \theta^{(n)}) = \sum_i \sum_{\omega_i: \text{finite}} \log(P((X_i, Y_i) = (x_i, y_i), W_{3i} = \omega_i | \theta)) P(W_{3i} = \omega_i | (X_i, Y_i) = (x_i, y_i), \theta^{(n)})$$

Define

$$s_i := E[W_{3i} | X_i = x_i, Y_i = y_i, \theta^{(n)}] = \sum_{\omega=0}^{\min(x_i, y_i)} \omega p_i(\omega | \theta^{(n)})$$

Then we find (ignore the constant)

$$Q(\theta | \theta^{(n)}) = \sum_i (x_i - s_i) \log \lambda_{1i} - \lambda_{1i} + (y_i - s_i) \log \lambda_{2i} - \lambda_{2i} + s_i \log \lambda_{3i} - \lambda_{3i}$$

with identifiable conditions. Notice Q can be separated into $Q(\lambda_1, \lambda_2)$ and $Q(\lambda_3)$. The two have no common parameters.

2 Q section

Considering the identifiable conditions we can either bring the constraints ($\sum_j$: except the last $att_j = -att_{\text{last}}$) into these $\log \lambda$, or optimize Q with constraints. We choose the latter.